

Syntax Analysis

Bottom Up Parsing

Bottom-Up Parsing

Bottom-up parsing parses the input string from the lexical analyzer. And if there is no error in the input string it constructs a parse tree as output. The bottom-up parsing starts building parse trees from the leaf nodes i.e., from the bottom of the tree. And gradually proceeds upwards to the root of the parse tree.

The bottom-up parsers are created for the largest class of **LR grammars**. As the bottom-up parsing corresponds to the process of reducing the string to the starting symbol of the grammar.

Step of Reducing a String:

1. A specific substring from the input string is identified.
2. A non-terminal in a grammar whose production body matches the substring is identified.
3. The substring in the input string is replaced by the non-terminal identified in step 2.

The main problem of bottom-up parsing is to decide:

- When to reduce which substring from the input string.
- Which production from the grammar must be applied.
- The reduction process in the bottom Up parsing is just the right most derivation in reverse order

Example

Productions:

$E' \rightarrow E$

$E \rightarrow E + T$

$E \rightarrow T$

$T \rightarrow T * F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow \text{id}$

States

$I_0: E' \rightarrow .E$

$E \rightarrow .E+T$

$E \rightarrow .T$

$T \rightarrow .T^*F$

$T \rightarrow .F$

$F \rightarrow .(E)$

$F \rightarrow .id$

$I_1: \text{goto}(I_0, E)$

$E' \rightarrow E.$

$E \rightarrow E.+T$

$I_2: \text{goto}(I_0, T)$

$E \rightarrow T.$

$T \rightarrow T.^*F$

$I_3: \text{goto}(I_0, F)$

$T \rightarrow F.$

$I_4: \text{goto}(I_0, ()$

$F \rightarrow (.E)$

$E \rightarrow .E+T$

$E \rightarrow .T$

$T \rightarrow .T^*F$

$T \rightarrow .F$

$F \rightarrow .(E)$

$F \rightarrow .id$

$I_5: \text{goto}(I_0, id)$

$F \rightarrow id.$

$T \rightarrow T.^*F$

$I_6: \text{goto}(I_1, +)$

$E \rightarrow E+.T$

$T \rightarrow .T^*F$

$T \rightarrow .F$

$F \rightarrow .(E)$

$F \rightarrow .id$

$I_7: \text{goto}(I_2, ^*)$

$T \rightarrow T.^*F$

$F \rightarrow .(E)$

$F \rightarrow .id$

$I_8: \text{goto}(I_4, E)$

$F \rightarrow (E.)$

$E \rightarrow E.+T$

$I_9: \text{goto}(I_6, T)$

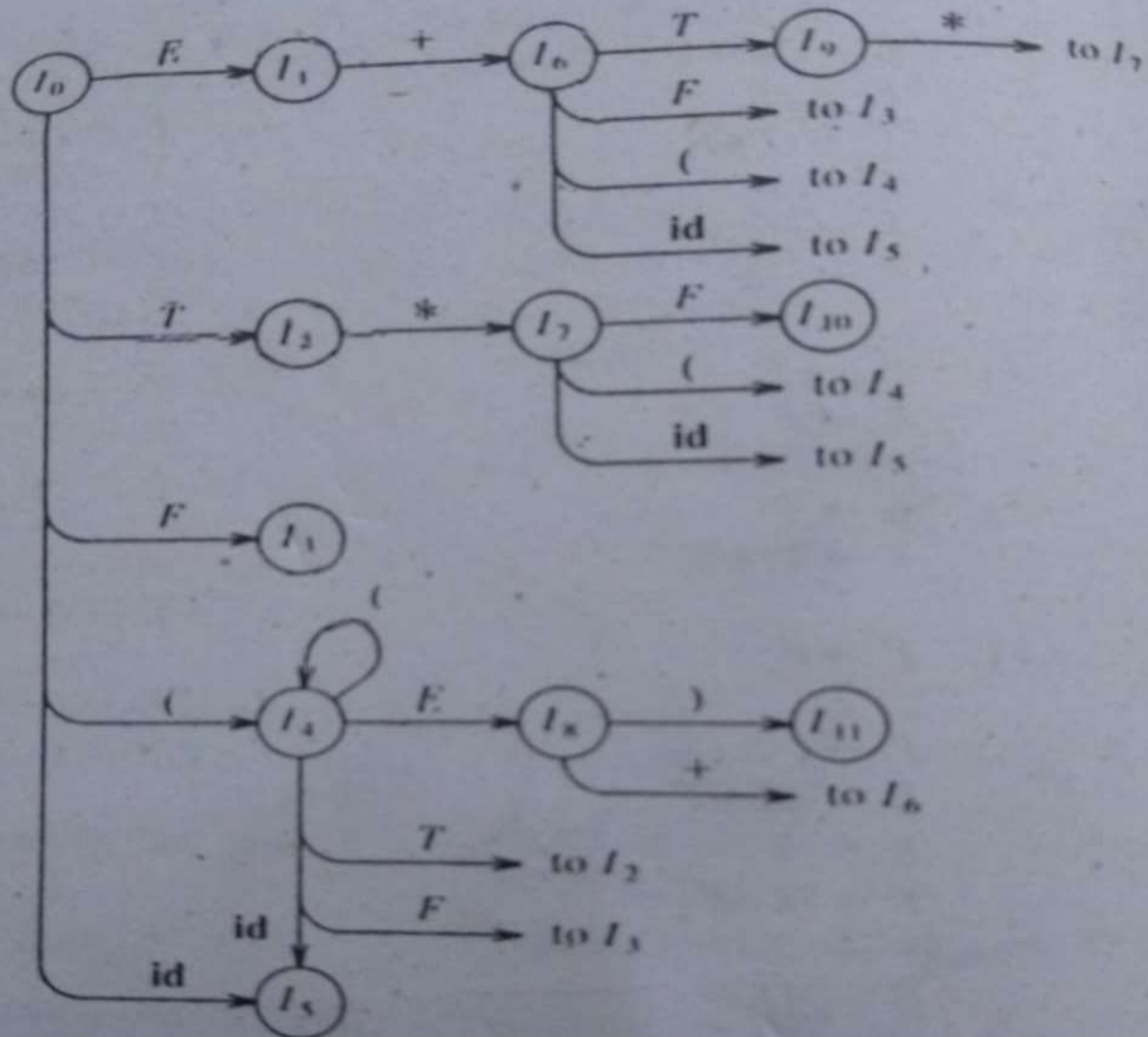
$E \rightarrow E+T.$

$I_{10}: \text{goto}(I_7, F)$

$T \rightarrow T^*F.$

$I_{11}: \text{goto}(I_8,))$

$F \rightarrow (E).$



Parsing Tables of Expression Grammar

Action Table

Goto Table

state	id	+	*	(	)	\$		E	T	F
0	s5			s4				1	2	3
1		s6				acc				
2		r2	s7		r2	r2				
3		r4	r4		r4	r4				
4	s5			s4				8	2	3
5		r6	r6		r6	r6				
6	s5			s4					9	3
7	s5			s4						10
8		s6			s11					
9		r1	s7		r1	r1				
10		r3	r3		r3	r3				
11		r5	r5		r5	r5				